

Simulated leaching of PFAS from land-applied municipal biosolids at agricultural sites

Jeff A.K. Silva^{a,*}, Jennifer L. Guelfo^b, Jiří Šimůnek^c, John E. McCray^d

^a Arclight Research & Consulting, Golden, CO, USA

^b Department of Civil, Environmental, and Construction Engineering, Texas Tech University, Lubbock, TX, United States of America

^c Department of Environmental Sciences, University of California, Riverside, United States of America

^d Civil & Environmental Engineering Department, Hydrologic Science & Engineering Program, Colorado School of Mines, Golden, CO, United States of America

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ABSTRACT

Biosolids are an important resource for agricultural practice but have recently received increased focus as a potential source of *per*- and polyfluoroalkyl substances (PFAS) in the environment. Few studies have investigated the transport of PFAS through the unsaturated zone under conditions relevant to biosolids application sites. Herein, the unsaturated flow and transport model HYDRUS is used to evaluate the leaching of *per*- and polyfluoroalkyl substances (PFAS) from land-applied biosolids used in agricultural practice to determine the impacts of PFAS leaching on underlying groundwater resources. This numerical case study was based on conditions and operations at two test sites in central Illinois where biosolids were applied at agronomic rates and where PFAS contents and desorption characteristics were previously characterized. Each site possessed different vadose zone soil textural heterogeneity. Simulations were performed under actual present-day meteorological conditions and extended 150 years beyond the initial biosolids application. These long-term simulations demonstrate how soil equilibrium sorption/desorption processes within the biosolids-amended surface soils effectively control the transport rate of individual PFAS to groundwater. Air-water interfacial (AWI) adsorption, which is sometimes considered to be a significant source of PFAS retention in vadose zone soils, was observed to have minimal impacts on PFAS leaching rates within the biosolids-amended surface soils at these sites. Additionally, the impact of AWI adsorption was found to be most significant for PFAS transport within the underlying vadose zone soils when these soils were more texturally homogeneous and considerably less significant within the texturally heterogeneous soils represented herein. The results of multiple long-term simulations were used to develop an empirical equation that relates predicted maximum PFAS pore-water concentrations reaching the saturated zone with changes in PFAS concentrations in the biosolids-amended soil for various biosolids re-application events. This approach is shown to be very useful in developing site-specific PFAS soil screening levels and/or maximum leachate levels for PFAS in support of establishing best management practices (BMPs) for land application of biosolids.

1. Introduction

Biosolids and products derived from biosolids are an important supplemental source of nutrients for agronomic, silviculture, and land restoration purposes. Biosolids are used in agriculture because they contain high concentrations of essential nutrients, primarily nitrogen and phosphorus, which are recycled into crop production to supplement or replace inorganic fertilizers. Biosolids also replenish soil organic matter, enhance soil water retention, and improve soil properties to support plant root growth. The total mass of biosolids applied to soils in

the United States (U.S.) is substantial; approximately 7.2 million dry metric tons of biosolids are generated in the U.S. annually, with roughly 55% (or 3.9 million tons) being land applied (Water Environment Federation WEF, 2017). While the agricultural application of biosolids represents a beneficial use of an otherwise discarded waste product, biosolids have also been shown to be a source of heavy metals, pharmaceuticals, and other environmental pollutants (e.g., USEPA, 1990; Kinney et al., 2006; Harrison et al., 2006; Clarke et al., 2010; Clarke and Smith, 2011; Higgins et al., 2011). This has fueled research aimed at determining the fate of these contaminants following land application

* Corresponding author.

E-mail address: jsilva@arclightrc.com (J.A.K. Silva).

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(e.g., Wilson et al., 1996; Chaney et al., 1996; USEPA, 2000; Yang et al., 2012; Yager et al., 2014) in support of best management practices for agronomic use.

Among the environmental pollutants associated with biosolids are *per*- and polyfluorinated substances (PFAS). PFAS are a group of environmental pollutants that have become ubiquitous in the environment due to their persistence and widespread use in consumer and industrial products and applications (e.g., USEPA, 2009; Ahrens and Bundschuh, 2014; Sharma et al., 2016). Recent research on the occurrence and fate of PFAS in biosolids has demonstrated that these chemicals can be persistent in soil (e.g., Washington et al., 2010; Sepulvado et al., 2011; Pepper et al., 2021), can be taken up by plants and edible crops (e.g., Higgins et al., 2011; Yoo et al., 2011; Blaine et al., 2013; Wen et al., 2016), and can leach from the soil and potentially impact underlying groundwater resources (Sepulvado et al., 2011). These findings have raised recent concerns about environmental and human health impacts associated with PFAS in land-applied biosolids. Such concerns have led to regulations of PFAS levels in biosolids in states such as Maine and Massachusetts, which require quarterly monitoring of PFAS in biosolids destined for land applications (Burns, 2019; Massachusetts Department of Environmental Protection (MassDEP), 2022).

Many PFAS are surface-active agents (or surfactants) with the potential to accumulate at environmental interfaces in the vadose zone environment. When released into the vadose zone soils, primary interfaces include air-water interfaces (AWI) within the water-unsaturated soil matrix and soil-water interfaces where PFAS can additionally associate with the solid phase (i.e., sorption). Both AWI adsorption and solid phase sorption combine to slow the rate of PFAS transport. The significance of PFAS adsorption to AWIs is dependent on the magnitude of the AWI area (A_{aw}) within the vadose zone, the surface activity of the PFAS, and the concentration of PFAS in the solution phase. The magnitude of A_{aw} within the vadose zone changes temporally and spatially within the vadose zone environment due to changes in soil moisture content (θ_w) as infiltrating rainwater moves vertically downward through the vadose zone. The surface activity of a specific PFAS molecule varies in response to its structural characteristics, such as the PFAS alkyl chain length (i.e., carbon number (C_N)) and functional group. Further, the amount of adsorbed PFAS depends on both concentration and solution ionic strength (Silva et al., 2019). PFAS sorption to model sands and natural soils has shown to be generally non-ideal (e.g., moderately non-linear and kinetically or diffusion controlled) (Guelfo et al., 2020; Brusseau, 2020). These and other complexities associated with PFAS vadose zone transport highlight challenges in predicting the environmental impacts due to the occurrence of PFAS in land-applied biosolids.

Standards of practice for land application of biosolids are based on the United States Environmental Protection Agency (USEPA) Part 503 Rule (i.e., 40 CFR Part 503, USEPA, 1993), which provides guidance for the agronomic use of biosolids based primarily on nutrient loading limits and mitigating the leaching of toxic metals. Currently, Part 503 does not consider other pollutants like PFAS. Further, the current understanding of PFAS leaching from land-applied biosolids, including the potential to impact underlying groundwater resources, is limited to field studies, in which surficial and depth-discrete soil samples have been collected and analyzed for PFAS over time (e.g., Washington et al., 2010; Sepulvado et al., 2011; Pepper et al., 2021). While these studies provide important information about the impacts of past biosolids application, the data collected typically represent snapshots of the vertical distribution of PFAS (i.e., type and mass distribution) resulting from PFAS leaching and do not provide much information needed to evaluate historical and future changes in the vertical distribution of PFAS. Prior field studies also have not provided data needed to evaluate important causal factors or environmental impacts, such as the effects of PFAS mass application rates associated with biosolids application rates, the rate of PFAS transport within the vadose zone, or the resulting impact on groundwater resources. To perform these evaluations, numerical methods

present a useful tool that can account for the dynamic temporal interchange of PFAS mass within the vadose zone in response to transient unsaturated flow conditions resulting from patterns of natural infiltrating precipitation.

The goal of this work was to evaluate the significance of land-applied biosolids as a long-term source of PFAS in agricultural applications under natural flow conditions, and to assess the potential for PFAS leaching from these surficial sources to underlying groundwater. Data collected from two field sites located in central Illinois was used to parameterize a version of the well-known HYDRUS unsaturated flow and transport model (Šimůnek et al., 2016) that was recently modified to simulate the transport of PFAS in the vadose zone (Silva et al., 2020). Here, simulated operational variables include vadose zone hydrostratigraphy, biosolids loading rate (BLR), and a number of repeated annual biosolids applications at a given BLR. Transport-specific variables and conditions were additionally included. As will be demonstrated, the numerical model was used to evaluate PFAS leaching from a source zone and vertical transport in the underlying vadose zone over extended periods, which in turn allowed for observation of the true timescales for PFAS transport from the surficial source to the underlying groundwater table.

2. Summary of site characteristics and model preparation

A detailed description of the methods and data used to prepare and parameterize HYDRUS for unsaturated water flow and PFAS transport is provided in the Supplemental Information (SI). The SI also includes a detailed discussion of the hydrogeologic characteristics of the test sites used in this evaluation and biosolids application information pertinent to this work. The SI also provides details pertaining to the numerical modeling approach.

Briefly, the two test sites selected for this evaluation are located within agricultural districts in central Illinois and were a part of a previous study evaluating the impacts of biosolids application on soil fertility and corn yields (Metropolitan Water Reclamation District of Greater Chicago (MWRDGC), 2008; Higgins et al., 2011). At these sites, designated Sites K and W, Class B biosolids were applied to individual test plots and were incorporated into the topsoil via tilling. Biosolids loading rates (BLR) varied between individual plots and were applied over three consecutive years (Section S1.1.4.1).

Notably, the objective of this current work was not to mirror test plot operations for the purpose of making site-specific predictions of groundwater impacts resulting from test operations at this site. Rather, this study used available data from these sites to parameterize HYDRUS for general use in evaluating the potential impacts of PFAS leaching to groundwater when biosolids are applied. In some cases, simulations used BLRs in excess of those actually applied at the test sites. However, simulations were performed to be consistent with actual dates of initial biosolids application and re-application at these sites. Specific site data used included historical precipitation and evaporation data (Section S1.1.1; Fig. S1), site soil types, and hydrostratigraphic data (Section S1.1.3).

Individual one-dimensional (1D) simulation profiles were constructed for both test sites (i.e., using HYDRUS) that extended from the ground surface to the water table (i.e., ~3 m below ground surface (bgs) at both sites) and incorporated soil textural changes observed at both sites. Surficial topsoil and vadose zone soils beneath Site K were homogeneous deposits of silty fine sand of glacial origin (Fig. S2). The surficial soil at Site W was a loam, and the underlying soils consisted of layers of silty clay, sandy loam, and medium-grained sands (Fig. S2). Textural characteristics and unsaturated hydraulic parameters for the soils are provided in Table S1 and Fig. S3.

Much of the biosolids-specific PFAS data used here was derived from the work of Sepulvado et al. (2011), who examined the leaching of PFAS from soils collected from these same test sites. This includes a linear biosolids loading rate (BLR, Mg/ha) model used to parameterize PFAS

mass input in the simulator, i.e.:

$$S_{BS,PFAS} = m BLR \quad (1)$$

where $S_{BS,PFAS}$ is the biosolids-amended soil concentration (ng/g_{soil}) and m is the slope of the BLR- $S_{BS,PFAS}$ regression, originally provided by Sepulvado et al. (2011) (Table S2). Specific PFAS of interest in this work included five homologous perfluoroalkyl carboxylates (i.e., perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA)) and two perfluoroalkyl sulfonates (i.e., perfluorohexane sulfonate (PFHxS) and perfluorooctane sulfonate (PFOS)). The perfluorinated tails of these surfactants consist of linear carbon chains and can be described by their carbon number (C_N) (e.g., C_N for PFHxA/PFHxS = 6 and for PFOA/PFOS = 8).

PFAS leaching from the biosolids was modeled as a two-site kinetic process that assumed sorption and desorption of PFAS from soil occurs as both an instantaneous process (i.e., Type 1 sites) and as a slower, rate-limited process (i.e., Type 2 sites) where the fractions of Type 1 (F_1) and Type 2 (F_2) are used in the model (i.e., $F_1 + F_2 = 1$; Section S1.1.4.3). Desorption rate coefficients (k) used to govern the rate-limited release of PFAS from agricultural soils were derived from another field study (Washington et al., 2010) (Section S1.1.4.2; Table S3).

Representative soil sorption parameters for the vadose zone soils underlying the biosolids-amended topsoil were derived from published laboratory experiments performed by Guelfo and Higgins (2013) and Guelfo et al. (2020). Specifically, PFAS sorption to these vadose zone soils was represented using either a linear sorption model or a non-linear Freundlich model, depending on the results of the transport experiments (Section S1.1.4.3; Table S4). Sorption and desorption within vadose zone soils were additionally modeled as a rate-limited process using the same two-site model used for the biosolids-amended topsoil. Representative F -values and k -values for these soils (Table S5) were derived from the work of Guelfo et al. (2020).

AWI adsorption was modeled as a Langmuir-type adsorption process and the dependence of AWI area on changes in soil moisture content (θ_w) was modeled using the Leverett thermodynamic model, also described in the SI (Section 1.1.4.4) and in the recent literature (Silva et al., 2020, 2022). The relationship between A_{aw} and θ_w used for the site soils is shown in Fig. S3; AWI adsorption parameters used in these simulations are provided in Table S6.

Finally, simulations used two monitoring points at both sites. The first was positioned at the base of the tilled topsoil containing biosolids (i.e., 30 cm bgs), which also represents the base of the PFAS source zone, to monitor the dynamics of PFAS leaching from the source zone. The second monitoring point was positioned at the top of the capillary fringe, representing the first point of contact with the saturated zone. This point provides information about the potential impacts of PFAS leaching from biosolids sources on groundwater. Pore-water in the capillary fringe is tension-saturated (i.e., hydraulic head is less than atmospheric pressure, or $h < 0$, where reference pressure is defined as 0 at atmospheric pressure). However, because it is fully saturated, significant horizontal flow first occurs here, which exchanges with flowing groundwater at and below the water table (defined as $h = 0$) (Silliman et al., 2002).

3. Results and discussion

3.1. Source zone dynamics

Generally, the magnitude and rate of loss of PFAS mass from the source ultimately govern the degree and longevity of the impact on underlying groundwater resources. Therefore, our first objective was to focus this evaluation on PFAS leaching from the biosolids applied to the surface soils. Simulations were performed for a single biosolids application event occurring in the Fall of 2004 for Site W and the Spring of 2005 for Site K. This timing of the simulated application events is in

keeping with the actual initial biosolids application for these sites. The simulated leaching of PFAS from the applied biosolids is characteristic of a temporally diminishing source (Fig. 1), with the rate of change in PFAS concentrations in the leachate (C_w) consistent with the comparative magnitude of the desorption coefficient (K_{des}) assigned to individual PFAS (Table S3). That is, PFAS with lower C_N leach from the biosolids source faster than those with higher C_N . This is consistent with field data where existing sources show more rapid depletion of lower C_N (i.e., lower molecular weight) PFAS (Sepulvado et al., 2011; Bekele et al., 2020).

The variability in C_w superimposed on PFAS leaching trends observed for some PFAS demonstrates the response of the solution-phase concentration to AWI adsorption as infiltrating precipitation changed the θ_w and magnitude of the A_{aw} throughout the simulation period. Note that increasing θ_w corresponds with decreasing A_{aw} , and vice versa. This variability is most pronounced for PFNA and PFDA and less pronounced for the remaining PFAS. This reflects the variability in the significance of AWI adsorption vs. interaction with the soil for a given PFAS relative to their interaction with the soil. The lower C_N PFAS (e.g., PFHxA, PFHpA, and PFHxS) are less surface-active, and therefore AWI adsorption is less significant. PFNA and PFDA are among the most surface-active solutes considered here (note the magnitudes of the AWI adsorption coefficients provided in Table S6). Therefore, in these cases, AWI adsorption competes more effectively with soil sorption processes to raise and lower C_w when θ_w and A_{aw} change within the PFAS source. In contrast, while PFDA and PFOS share similar surface activity, the temporal trend in C_w for PFOS does not exhibit the same degree of variability observed for PFDA. PFOS, in this case, is more strongly associated with the soil phase (i.e., highest K_{des} ; Table S3) which out-competes AWI adsorption in the distribution of PFAS mass.

Generally, rates of PFAS desorption from the source zones for the lower C_N PFAS were observed to increase with decreasing BLR over the 15-year simulation period (Figs. S4 and S5). However, for the higher C_N PFASs, the rate of decrease in concentration was much less dependent on the BLR (Figs. S4 and S5). These results are attributed to differences in the affinity of the different PFAS for the biosolids-amended soil as sorption/desorption processes within the source soils retard the transport of the lower C_N PFASs than the higher C_N PFASs. Additionally, for a given BLR, the rate of PFAS leaching was found to be largely insensitive to the rate constant used (Figs. S4 and S5) even when its value is varied by two orders of magnitude (Table S6). This lack of sensitivity is attributed to (1) the generally high K_{des} of the biosolids-amended soils, (2) the relatively small fraction of rate-limited sites associated with these same soils ($F_2 = 0.16$ in this case), and (3) the low concentration gradients associated with low PFAS leachate concentrations, which limit the driving force for mass transfer to the rate-limited sites. The limited influence of sorption/desorption kinetics on the leaching of PFAS from shallow source zones for observed for the 15-year simulations performed here is consistent with a prior long-term leaching evaluation (Kuntz and Grathwohl, 2009). These results demonstrate that equilibrium sorption would be an appropriate transport model simplification for Sites K and W. This result is significant because it suggests that when biosolids are applied to surface soils at typical agronomic loading rates, predicted rates of PFAS mass loss from the surficial soil source are more dependent on an accurate characterization of the equilibrium sorption/desorption parameters than on non-equilibrium parameters.

3.2. Vadose zone hydrodynamics

Vadose zone hydrodynamics are indicative of the interplay between changes in soil moisture and other soil hydraulic and physical properties. Infiltrating precipitation increases θ_w , resulting in a corresponding increase in unsaturated hydraulic conductivity (K_{unsat}), the downward flux of water (q ; the volume of water per unit area per time) and contaminant mass. Changes in θ_w vary spatially and temporally within the vadose zone resulting in the transient character of the hydraulic

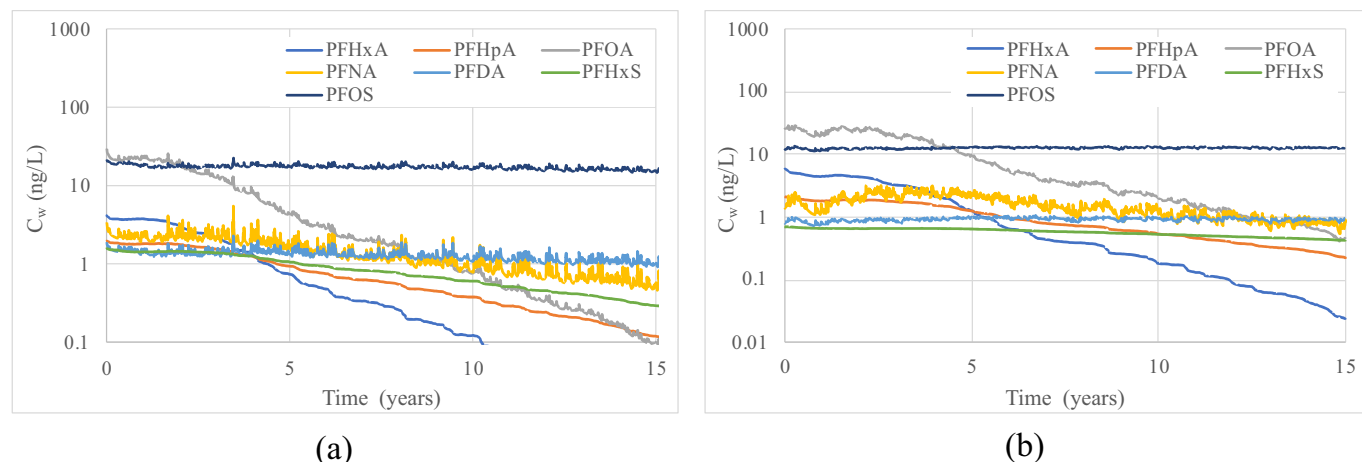


Fig. 1. Example of simulated PFAS release from both test sites: a) Site K and b) Site W. In both cases, BLR = 5 Mg/ha. Note that year 15 in these plots is the calendar year 2020.

properties profiles (Fig. 2). Similarly, changes in θ_w result in concomitant changes in A_{aw} (e.g., Silva et al., 2022), which governs the amount of PFAS mass adsorbed at AWIs, and thereby regulates the mass flux of contaminants within the vadose zone.

For Site K, the PFAS source zone and the underlying vadose zone share the same hydraulic properties but different transport characteristics (i.e., soil sorption/desorption coefficients and rate constants). Hydraulically, Site K is modeled as a homogeneous system, and the response observed in the parameters in Fig. 2 reflects this homogeneity. K_{unsat} values are periodically very low near the soil surface due to low θ_w due to evaporation. However, K_{unsat} generally ranges between 0.01 and 1 cm/day below the surface. This is well below the saturated conductivity (21.2 cm/day) of this silty/sandy soil, and q is thus more variable in this system and more dependent on precipitation variability. This well-drained soil provides a lower pseudo-steady state moisture content that yields elevated A_{aw} values (ranging between 200 and 400 cm²/cm³) and promotes more significant retention via AWI adsorption in the upper vadose zone relative to Site W (see below).

The vadose zone at Site W is texturally and hydraulically heterogeneous. The loamy PFAS source zone retains more soil moisture during the simulation period, which is shown in the uppermost layer of Fig. 2g. Generally, θ_w is shown to increase in time within individual strata, resulting in independent inter-layer changes in A_{aw} values. The general increase in retained moisture increases K_{unsat} values within the system, resulting in a reasonably uniform K_{unsat} distribution vertically. Due to the higher overall θ_w , K_{unsat} in Site W soils can more readily approach the saturated hydraulic conductivity. The pseudo-steady state flux of water in the Site W vadose zone is also generally vertically uniform, although the water flux is still lower in the Site W vadose zone than in the Site K vadose zone. However, incidents of greater than average rainfall can significantly increase q in this system (i.e., note the significant increase in q at Year 15; Fig. 2f). Given the hydraulic and sorption heterogeneity, PFAS would be expected to transport more slowly in the vadose zone at Site W vs. Site K, and the transport rate will be less sharply affected by precipitation patterns. Leaching of PFAS from the source zone would also be expected to be slowed due to the hydraulic limitations of the underlying strata.

3.3. Significance of PFAS AWI adsorption

AWI adsorption can play a significant role in regulating the rate of PFAS transport in the vadose zone. However, this was not the case within the biosolids-amended source zones of Sites K and W. Regardless of whether AWI adsorption was or was not considered in these

simulations, the overall rate of PFAS leaching from the biosolids-amended soil source zone into the vadose zone does not change at either site (Fig. 3). These results again demonstrate the dominance of soil sorption/desorption processes on PFAS leaching from the surficial source soils in these simulated cases. Note, again, the fluctuating C_w values in simulations that considered AWI adsorption compared to smooth changes in C_w values when AWI adsorption was neglected. This is because AWI adsorption depends on changes in θ_w and AWI area, while soil sorption does not. Infiltrating water increases θ_w and reduces AWI area, producing the release of PFAS from AWI to the soil solution. On the other hand, evaporation decreases θ_w and increases AWI area, producing the mass transfer of PFAS from the soil solution to AWI.

The effect of AWI adsorption on the simulated transport of PFAS within the entire vadose zone is shown in Figs. 4 and 5 for Sites K and W, respectively. For the less surface-active PFAS, AWI adsorption does not significantly contribute to transport at either site. However, as the surface activity of the solute increases, AWI adsorption plays a more significant role in the retention of these solutes. For example, at the more homogeneous Site K, transport of PFOA is observed to be moderately accelerated, and the transport of PFOS is significantly accelerated when AWI adsorption is not considered in the model. The effect is not nearly as contributive at the texturally heterogeneous Site W. While clear differences exist in the rate of change of PFAS concentrations in the upper part of the vadose zone, after 15 years, PFOA concentrations contacting the saturated zone are not significantly different whether AWI adsorption is considered or not. Likewise, after 15 simulated years, PFOS at Site W has not contacted the saturated zone and has transported to essentially the same depth (i.e., ~265 cm) regardless of whether AWI adsorption is considered.

Assuming equivalent meteorological inputs, the significance of AWI adsorption on the overall transport of surface-active solutes is dependent on three primary factors: 1) the degree of affinity for the solid phase (as demonstrated for the biosolids amended soils), 2) the surface activity of the solute (i.e., an affinity for the AWI), and 3) the magnitude of A_{aw} as it changes during infiltration events (i.e., higher A_{aw} values associated with lower water content values provide increased AWI adsorption and retention of surface-active solutes). Factor 3 is strongly influenced by textural heterogeneity. At Site K, the modeled soil textural homogeneity of the vadose zone soils allows for the establishment of a smaller pseudo-steady state θ_w condition that promotes AWI adsorption in the upper vadose zone (Fig. 2) by maintaining higher A_{aw} . In the absence of AWI adsorption, transport occurs much faster because transport is now dependent only on the hydraulic properties of the media and interactions with the solid phase. At Site W, the modeled soil textural

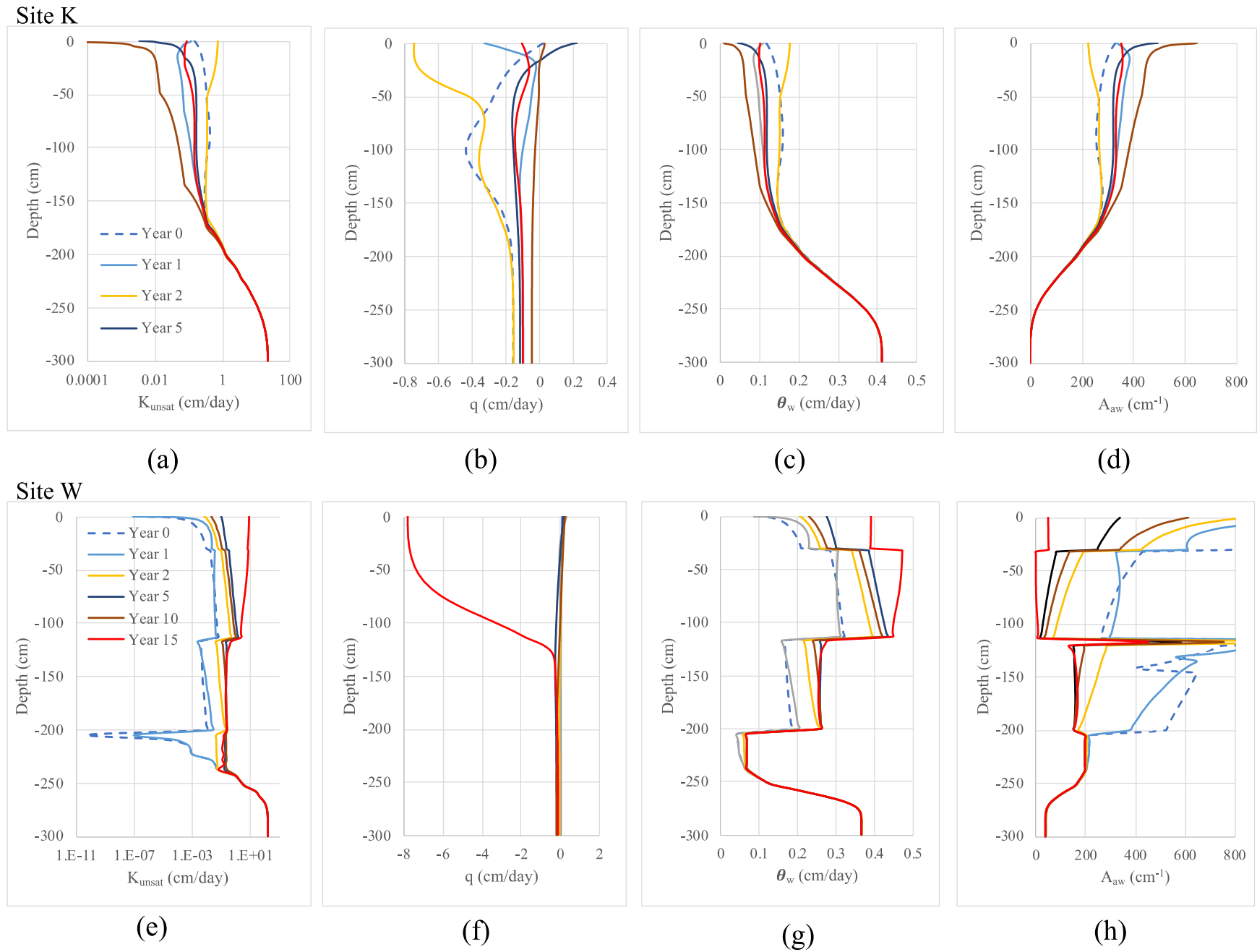


Fig. 2. Simulated variability in unsaturated hydraulic conductivity, water flux, moisture content, and air-water interfacial area for Site K (a-d) and Site W (e-h). Fluxes are positive upward and negative downward.

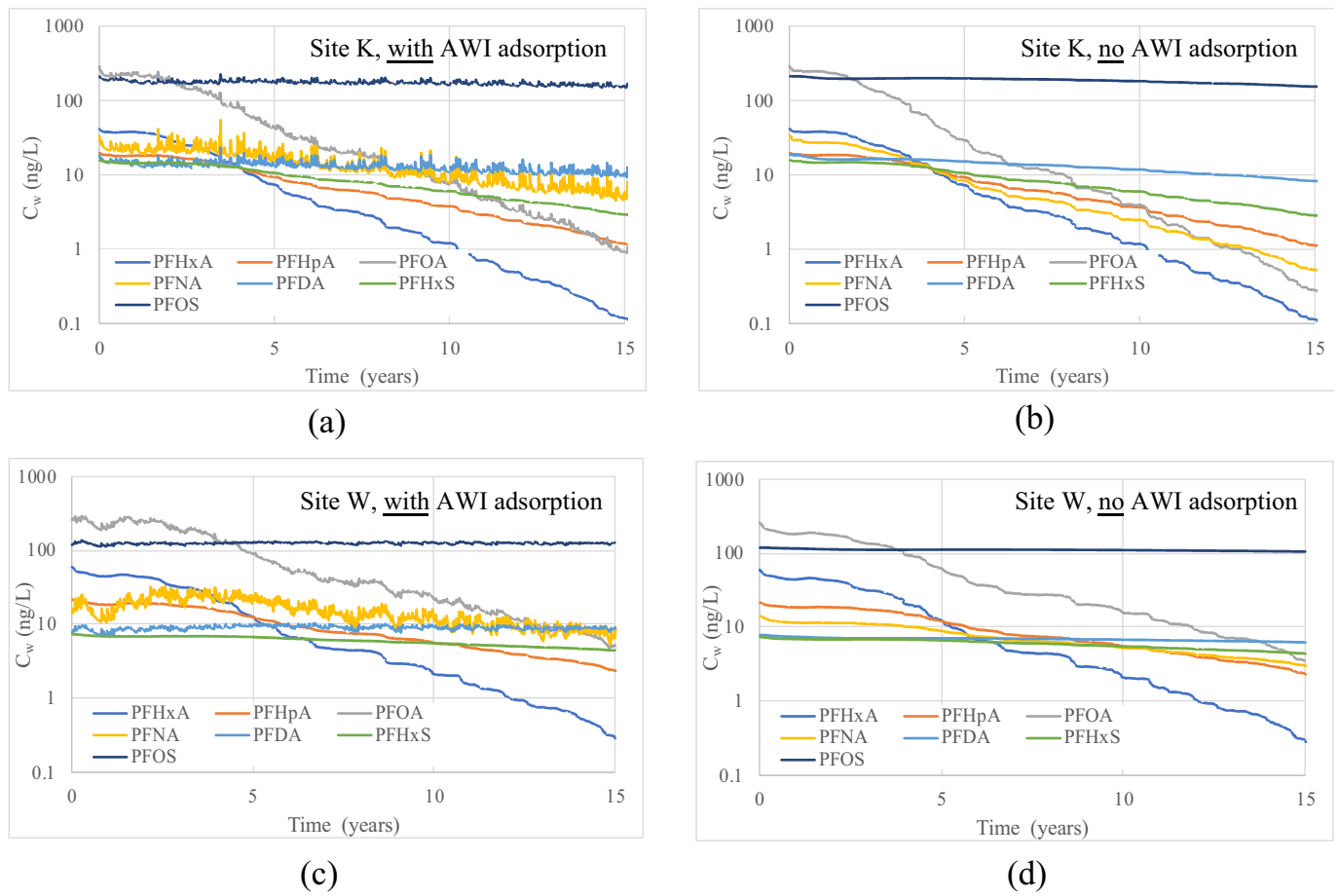


Fig. 3. Simulated PFAS concentration profiles within the biosolids-amended source zone with and without AWI adsorption at Site K (a-b) and Site W (c-d). Rate constants were those reported by Washington et al. (2010), as presented in Table S3. BLR = 50 Mg/ha in all cases.

heterogeneity limits the contribution of AWI adsorption because the soils beneath the biosolids-amended source zone maintain a higher θ_w and, as a result, lower A_{aw} . AWI adsorption is therefore not significant in the upper Site W vadose zone and does not contribute significantly to the retention of PFAS in this system. At Site W, PFAS transport is dominated by interactions with the soils and the hydraulics of unsaturated flow.

These results demonstrate how soil textural heterogeneity can, under certain conditions, play a more important role than AWI adsorption in determining the magnitude and rate of PFAS transport in the vadose zone, even for more surface-active PFAS (e.g., PFOS in these cases). The downward coarsening sequence of soil layering present at Site W (Fig. S2) promotes this heterogeneity-imposed hydraulic control on PFAS transport. Alternatively, if the soils beneath the topsoil were coarser (e.g., sand), steady state soil moisture would likely be lower within that underlying coarse layer and A_{aw} values would be much higher (see $A_{aw}(\theta_w)$ curves, Fig. S3c), and AWI adsorption would play a greater role in PFAS retention. However, hydraulic conditions within the Site W loam topsoil would still limit the rate and magnitude of leaching in this case because of the intermittent release of PFAS from that source soil due to the capillary barrier effect (i.e., infiltrating precipitation would need to build up enough hydraulic head at the loam/sand interface to overcome the suction in that layer and initiate flow within the underlying sand). Further, the flux of water entering the coarser media would be limited by the overall lower hydraulic conductivity of the surface soil. Therefore, soil textural heterogeneity should be considered carefully when simulating PFAS leaching in the vadose zone.

3.4. Leaching of PFAS from a land-applied biosolids source

3.4.1. Single-event biosolids application

To better evaluate PFAS leaching in these systems, PFAS transport in the vadose zone was simulated for a single biosolids application over a longer simulated time period. An example of the simulated arrival of PFAS at the top of the capillary fringe at both test sites is presented in Fig. 6 for BLR = 50 Mg/ha. Immediately apparent in Fig. 6 are the differences in the time needed for these PFAS to reach the saturated zone. The less surface-active PFAS (i.e., PFHxA, PFHpA, PFOA, PFHxS) are generally less retained by soil interactions and AWI adsorption in these systems and therefore move more rapidly through the vadose zone. However, the higher C_N PFAS (i.e., PFNA, PFDA, and PFOS in this case) are more surface-active and sorbing, and are considerably more retained, requiring many decades to transport to the saturated zone and impact groundwater resources. The times required to achieve maximum PFAS concentrations are also remarkable, with PFOS shown to require ~60 years to reach a peak concentration at the top of the saturated zone at Site K and ~109 years at Site W.

Cumulative mass flux (CMF) plots additionally demonstrate the differences in arrival times for PFAS represented in this work (Fig. 6). CMF values are presented using hectares as the areal dimension to be consistent with units used for the BLRs and are useful as input datasets (i.e., source terms) for groundwater (i.e., saturated zone) modeling efforts. The time to approach maximal CMF reflects the transport characteristics of the individual PFAS, and the magnitude of the maximum CMF achieved reflects the initial PFAS mass loading within the source zone for a given BLR. For these sites, the maximal cumulative PFAS mass flux

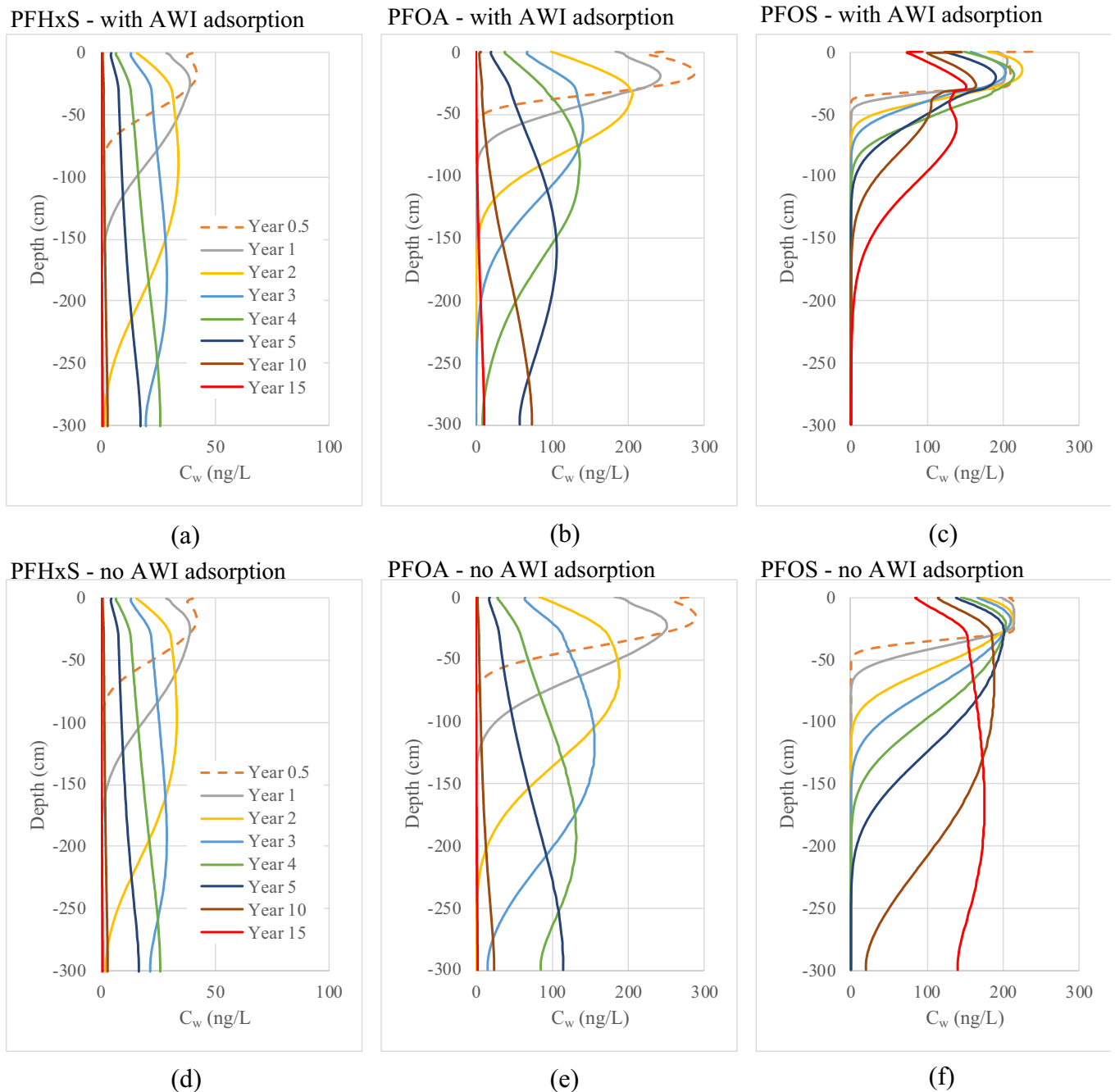


Fig. 4. PFAS concentrations in pore-water transported to the groundwater table during the first 15 years since biosolids were applied at Site K (BLR = 50 Mg/ha) simulated with AWI adsorption (a-c) and without AWI adsorption (d-f). Year 15 in these plots is the calendar year 2020.

entering the saturated zone was 1 to 100 g per hectare of biosolids-amended land but, in some cases, required several decades to reach that loading.

Expanding this evaluation, simulations were performed for additional single event biosolids applications at BLRs of 5, 10, 25, 50, and 100 Mg/ha. The results of these simulations for PFOA and PFOS are presented in Fig. 7. Simulated maximum or “peak” leachate concentrations reaching the saturated zone ($C_{w, \max}$) for all PFAS evaluated herein (Table S7) increased linearly with increasing BLR for all PFAS evaluated, consistent with the use of a linear loading rate model used to parameterize PFAS mass input in the simulator (Eq. (1)).

The slope of the BLR- $C_{w, \max}$ regression can be used to directly estimate the maximum BLR for a given site and PFAS. However, because the

PFAS mass associated with the applied biosolids can change from source to source and over time, it is more practical to represent predicted $C_{w, \max}$ values as a function of $S_{BS, PFAS}$ (e.g., PFOA and PFOS in Fig. 8). $S_{BS, PFAS}$ values were calculated using Eq. (1), and the values for all PFAS included in this evaluation are presented in Table S8. The slopes of the linear regressions of this data are likewise provided in Table S9. These data can then be used to directly estimate $C_{w, \max}$ values for these sites and simulated conditions if $S_{BS, PFAS}$ is determined for a given biosolids application.

3.4.2. Multiple biosolids applications

In practice, biosolids are often applied annually over multi-year cycles, as was the case for test Sites K and W (Higgins et al., 2011).

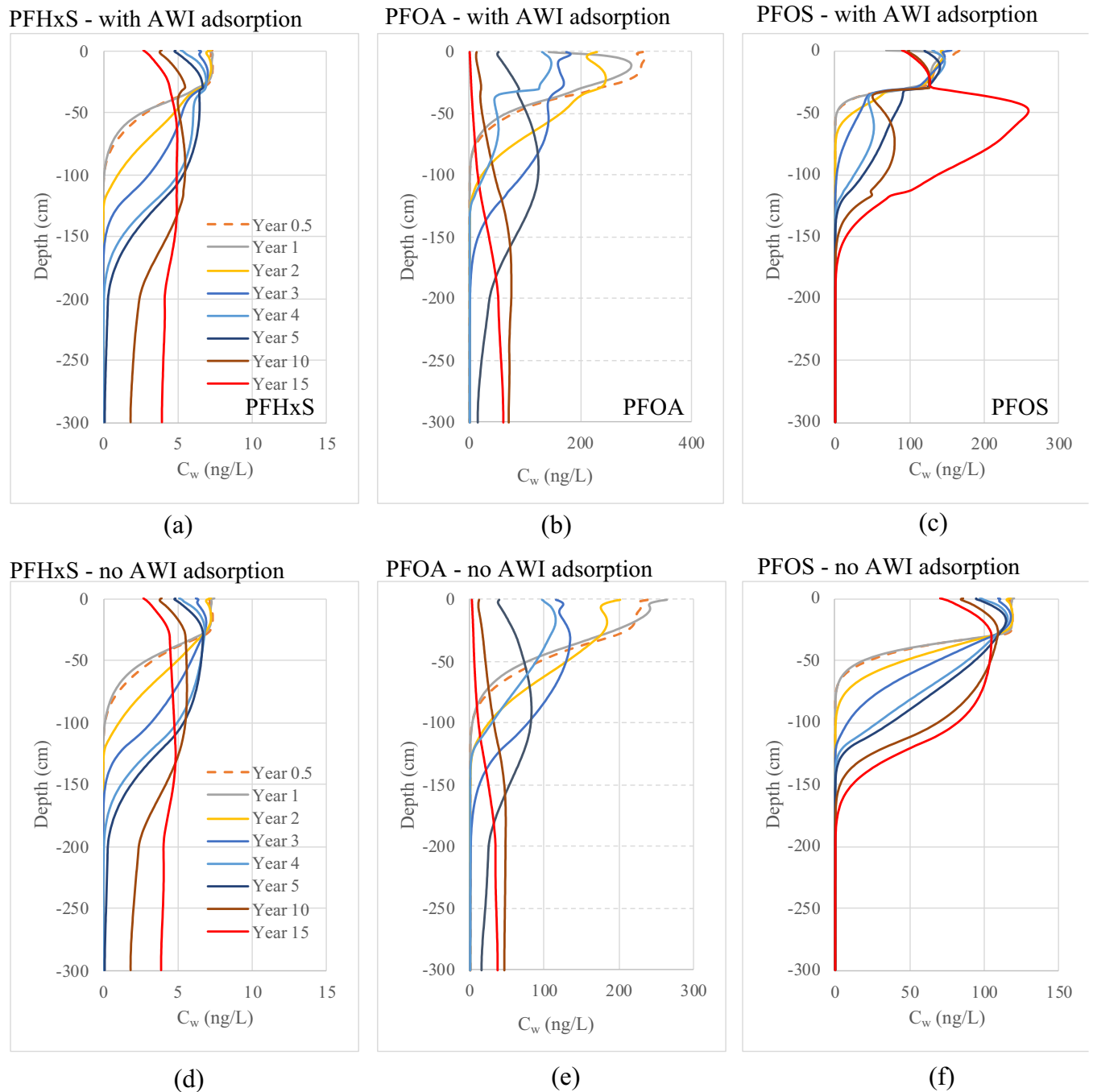


Fig. 5. PFAS concentrations in pore-water transported to the groundwater table during the first 15 years since biosolids were applied at Site W (BLR = 50 Mg/ha) simulated with AWI adsorption:(a-c) and without AWI adsorption (d-f). Year 15 in these plots is the calendar year 2020.

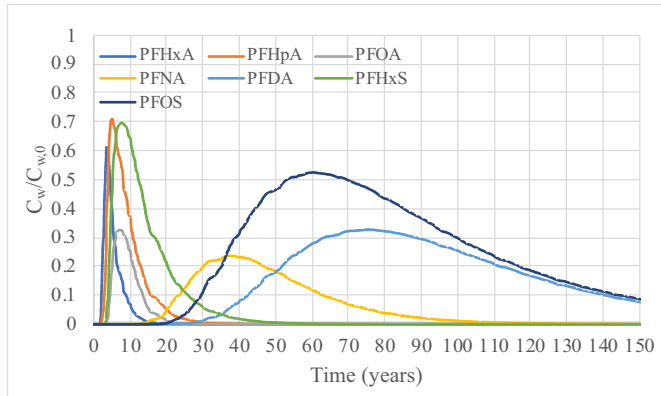
Therefore, additional simulations were performed to examine the effect of multi-year biosolids applications on maximum PFAS concentrations in leachate reaching the saturated zone. Lower and more agronomic BLR's of 5, 10, and 25 Mg/ha, and annual biosolids re-application periods (T_{app}) of 1, 3, 5, and 10 years were used in these simulations. The remaining conditions and parameters were the same as in previous simulations. Examples of the simulated arrival of PFOA and PFOS at different application periods are presented in Fig. 9 for both test sites. Additionally, simulated $C_{w, max}$ values for PFOA and PFOS resulting from the various application periods and loading rates are provided in Fig. 10.

As shown in Fig. 10, simulated $C_{w, max}$ was observed to increase nonlinearly with T_{app} , and a 2nd-order polynomial regression equation

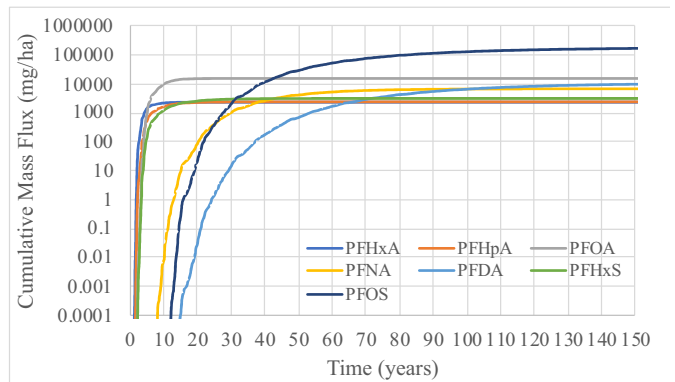
(forced through the origin) was used to represent the trend in the data. For the T_{app} - $C_{w, max}$ relationships, the degree of non-linearity increased with increasing BLR, which is most clearly observed in PFOA results (e.g., Fig. 10a and c) but is also evident in Fig. 9, where the leading edge of the PFOA arrival profiles are coincident, which indicates the initial PFAS arrival times are independent of T_{app} for a given BLR.

Similar to increasing the BLR, multiple annual biosolids applications increase the PFAS $C_{w, max}$ reaching the groundwater table. However, multiple annual applications ultimately provide a greater $C_{w, max}$ values for a given BLR due to the re-application of a fresh source of PFAS that increases the PFAS mass within the surface soils (i.e., $S_{BS, PFAS}$) and in the leachate. These results indicate a practical limit for biosolids

Site K

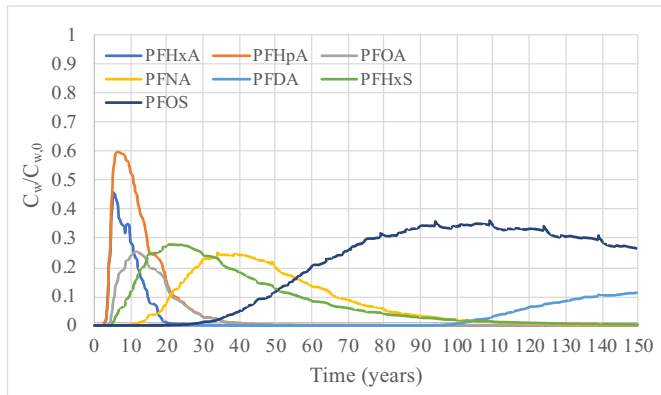


(a)

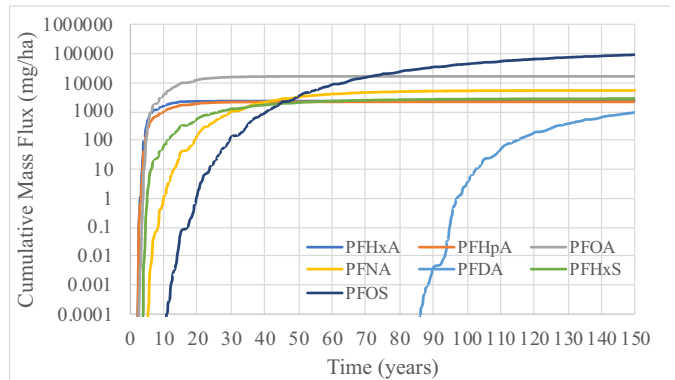


(b)

Site W



(c)



(d)

Fig. 6. Simulated PFAS transport within the Site K and Site W vadose zones for a single biosolids application with BLR = 50 Mg/ha: (a/c) PFAS relative concentrations at the capillary fringe monitoring location, and (b/d) cumulative mass flux of PFAS impacting saturated zone.

applications in agriculture would need to be derived to be protective of underlying groundwater resources.

Like regression equations presented in Figs. 7 and 8, those provided in Fig. 10 could also be used to estimate $C_{w, max}$ for PFAS at different values of T_{app} and for the BLR and associated $S_{BS, PFAS}$ values applied. However, a unifying predictive relationship is preferred to include the impacts of both $S_{BS, PFAS}$ and T_{app} for an individual site. To this end, the slope values of the linear $S_{BS, PFAS}-C_{w, max}$ regressions (Table S9) were regressed against T_{app} and fit using a 2nd-order polynomial regression equation (forced through the origin). These $m-T_{app}$ relationships were then used in combination with $S_{BS, PFAS}-C_{w, max}$ regression equations to derive the following polynomial expression that allows the prediction of $C_{w, max}$ for various combinations of $S_{BS, PFAS}$ and T_{app} as:

$$C_{w, max} = S_{BS, PFAS} \left(a_1 (T_{app})^2 + a_2 (T_{app}) \right) \quad (2)$$

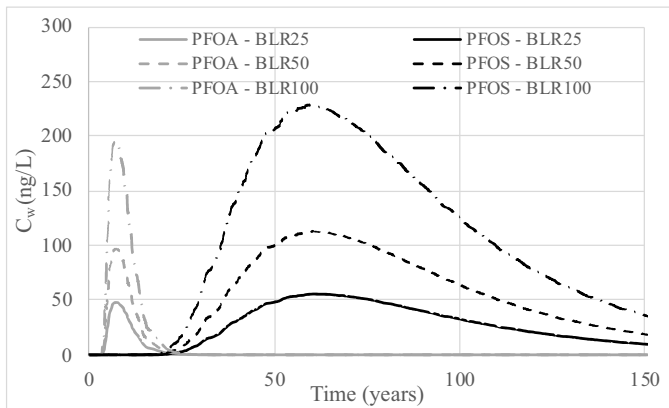
where a_1 and a_2 are regression coefficients for the PFAS of interest (Table 1). This approach was used for both sites. As proposed, Eq. (2) is site-specific and additionally dependent on the linearity of $S_{BS, PFAS}$ with changes in BLR for the biosolids applied. Therefore, additional model simulations would need to be performed for a given site to derive new site-specific a_1 and a_2 parameters. However, once derived, Eq. (2) could be applied to the development of site-specific PFAS soil screening levels and/or maximum leachate levels for PFAS for proposed $S_{BS, PFAS}$ and T_{app} to develop best management practices (BMPs) for land application of biosolids.

3.5. Implications for groundwater contamination

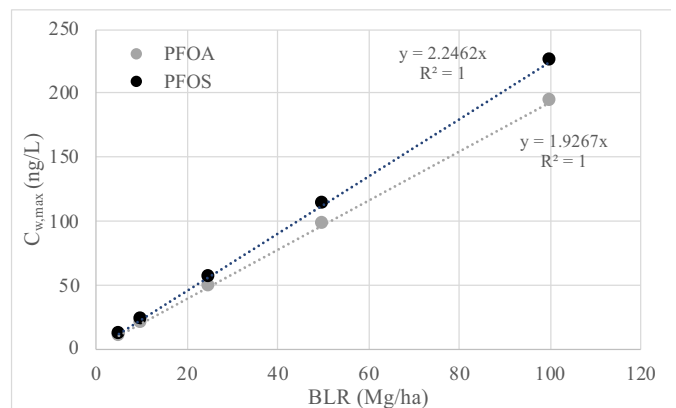
The USEPA recommended the use of 70 ng/L for PFOA and PFOS (i.e., individual or combined) as the preliminary remediation goal (PRG) for groundwater contaminated by PFAS, which is a current or potential source of drinking water for sites where state or local metrics have not been adopted (USEPA, 2019). If this same metric is used to define maximum allowable leaching levels for PFOA and PFOS, the allowable leaching level is shown to have been exceeded at Site K and Site W for a single BLR of 50 Mg/ha (Fig. 8). Likewise, this PRG is met or exceeded for several multi-year applications shown in Figs. 9 and 10.

However, more typically, the PRG of 70 ng/L would be for PFOA and PFOS arriving or produced at a down-gradient receptor point (e.g., a surface water body or groundwater well). Therefore, depending on the distance between the receptor and the biosolids application area, attenuation processes may allow PFAS leachate concentrations immediately beneath the biosolids application site to be in excess of 70 ng/L. Well-known analytical solutions for solute transport (e.g., Domenico, 1987) have been incorporated into various analytical models to allow the estimation of dilution attenuation factors (DAF) at user-defined distances from the source of contamination (e.g., Nevin et al., 1997; USEPA, 2000). Using the Domenico model for a sorbing and non-decaying solute (Section S3) and relevant site-specific inputs for these sites, DAF values were calculated for PFOA and PFOS at Site K (Table 2). The DAF value was used to determine $C_{w, max}$ for a 70 ng/L arrival at a given receptor distance from the source. In turn, these $C_{w, max}$ values were used to determine the corresponding $S_{BS, PFAS}$ values for PFOA and

Site K

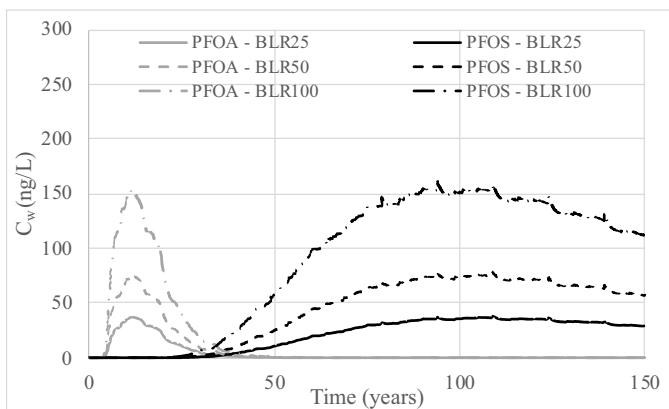


(a)

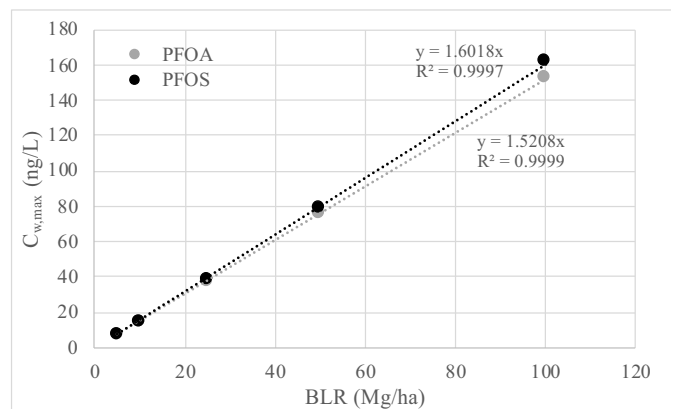


(b)

Site W

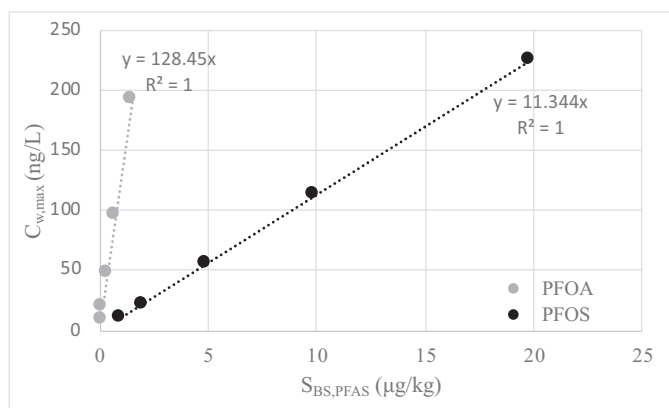


(c)

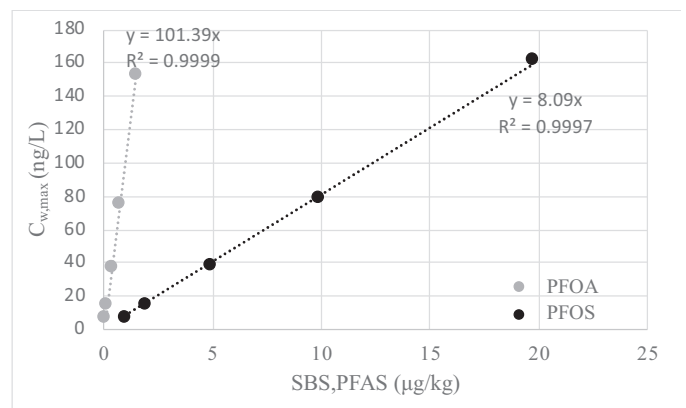


(d)

Fig. 7. Results of varied BLRs for a single application event: a/c) Simulated PFOA and PFOS arrival profiles within the Site K and Site W vadose zones, respectively, and b/d) maximum PFOA and PFOS concentrations contacting the saturated zone for Site K and Site W, respectively.



(a)



(b)

Fig. 8. Maximum PFOA and PFOS concentrations contacting the saturated zone at both sites: a) Site K and b) Site W.

PFOS using the linear regression and slope values described above (Table S9). These $S_{BS, PFAS}$ values would then be the maximum values, or potential site-specific soil screening levels (SSL) for PFOA and PFOS, that represent the maximum PFAS mass per unit mass of soil that could

be applied to the land surface either in a single event or accumulated or over multiple application events to avoid exceeding the 70 ng/L PRG at a distant receptor. For example, a receptor positioned 500 ft (ft) down-gradient of Site K would see PFOA and PFOS concentrations exceed the

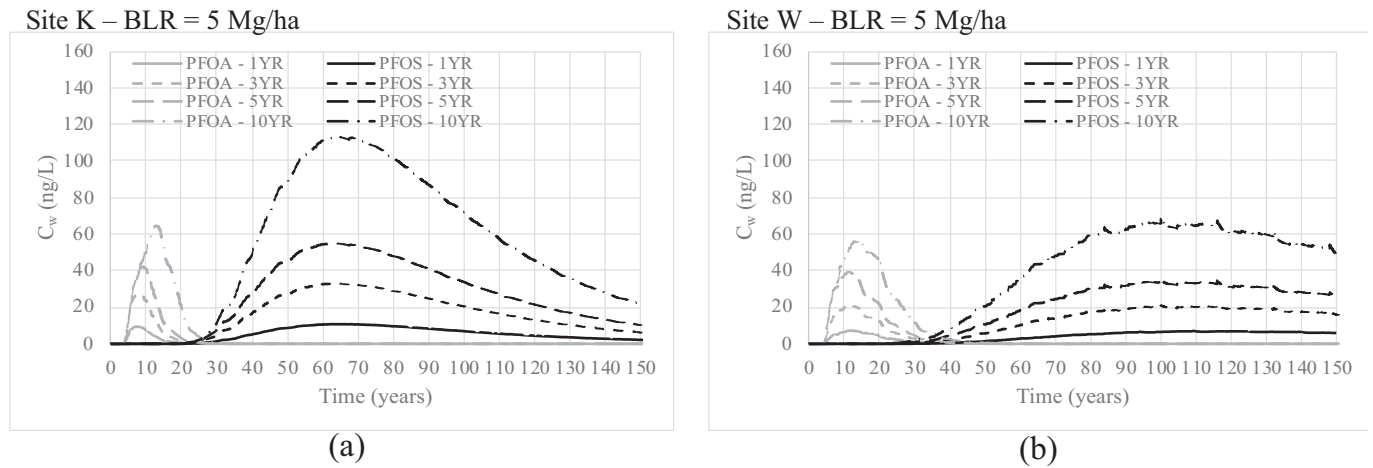


Fig. 9. Simulated PFOA and PFOS arrival profiles for different annual application periods at Site K(a) and Site W (b) for BLR = 5 Mg/ha.

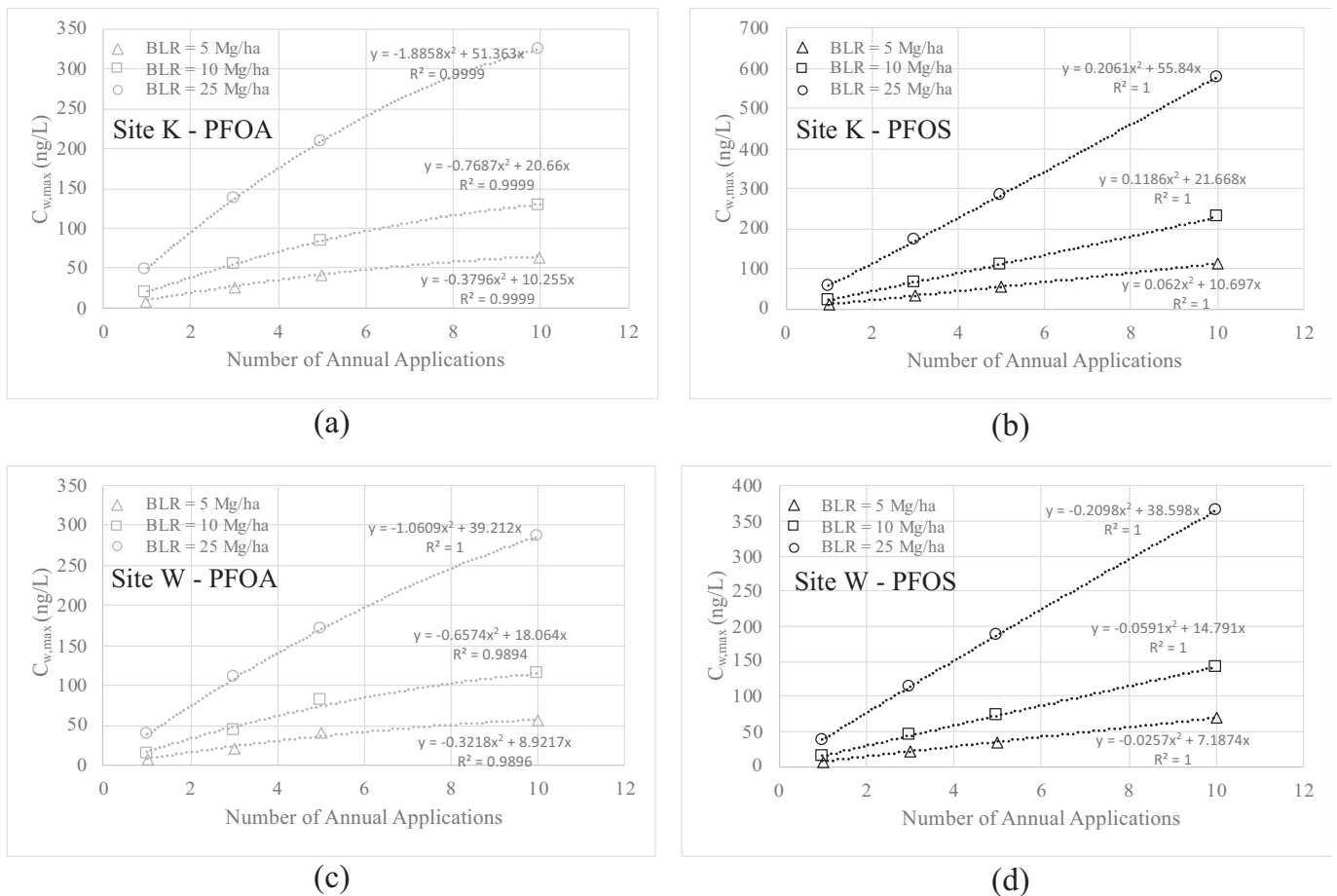


Fig. 10. Dependence of $C_{w, \max}$ values for PFOA and PFOS on the number of annual biosolids applications at Site K (a/b) and Site W (c/d).

70 ng/L limit when $S_{BS, PFOA}$ and $S_{BS, PFOS}$ at Site K exceed 4.32 and 48.9 $\mu\text{g/kg}$, respectively, or when the cumulative BLR in each case exceeds 288 and 247 Mg/ha, respectively. Given the significant amount of time needed for PFOA and PFOS transport in the vadose zone and the arrival of the peak concentration at the saturated zone, the potential for PRG exceedances at potential receptors should additionally be considered in both the nearer term (i.e., based on PFOA peak arrival times) and in the

much longer-term (i.e., based on peak PFOS peak arrival), as maximum solution concentrations at the receptors could occur many decades after initial biosolids application.

Regarding the continued and long-term use of biosolids at these sites, the simulated results of this work suggest biosolids application should be a strictly managed activity in a manner consistent with the management methods outlined in USEPA 40 CFR Part 503 for metals. The BLR values

Table 1

Results of multi-year biosolids application simulations and regression parameters for PFOA and PFOS used in developing and applying Eq. (2).

Site K		$C_{w,max}$ (ng/L) for different BLR			$S_{SB,PFAS}-C_{w,max}$ regression		$m-T_{app}$ regression		
PFAS	T_{app} (years)	BLR = 5 Mg/ha	BLR = 10 Mg/ha	BLR = 25 Mg/ha	m	R^2	a_1	a_2	R^2
PFOA	1	9.41	19.0	47.9	127.4	1	-5.04	137	1
	3	27.1	54.6	138	366.8	1			
	5	42.2	84.8	210	560	1			
	10	64.5	130	325	866.1	1			
PFOS	1	10.8	21.8	55.5	147.6	1	0.5915	148.1	1
	3	32.9	66.5	170	450.6	1			
	5	54.8	111	284	755	1			
	10	113	229	579	1540	1			

Site W		$C_{w,max}$ (ng/L) for different BLR			$S_{SB,PFAS}-C_{w,max}$ regression		$m-T_{app}$ regression		
PFAS	T_{app} (years)	BLR = 5 Mg/ha	BLR = 10 Mg/ha	BLR = 25 Mg/ha	m	R^2	a_1	a_2	R^2
PFOA	1	7.24	14.6	37.2	98.8	1	-3.09	107.2	1
	3	21.1	42.7	109	288	1			
	5	39.4	79.6	170	465	1			
	10	56.6	114	286	762	1			
PFOS	1	7.21	14.8	38.4	102	1	-0.53	102.1	1
	3	21.3	43.8	114	302	1			
	5	35.3	72.5	188	497	1			
	10	69.3	142	365	968	1			

Table 2

DAF and soil screening level calculations for PFOA and PFOS for site K.

Distance from source (ft)	DAF	$C_{w,max}$	$S_{SB,PFOA}$	$S_{SB,PFOS}$	PFOA BLR (Mg/ha)	PFOS BLR (Mg/ha)
1	1	70	0.54	6.17	36	31
10	1	70	0.54	6.17	36	31
100	1.46	102	0.80	9.01	53	45
250	3.25	228	1.77	20.1	118	101
500	7.92	554	4.32	48.9	288	247
750	15.6	1095	8.52	96.5	568	487
1000	26.4	1848	14.4	163	959	823
5280	691	48,370	377	4265	25,105	21,534

Note: $C_{w,max}$ are those calculated for 70 ng/L concentration at a given receptor distance from the source.

greater than those in Table 2 would result in exceedances of the 70 ng/L leachate level for PFAS at different receptor distances at Site K. However, given the longevity of biosolids as a source of PFAS and the significant times involved in PFAS contamination of groundwater for these sites, the same BLR/ $S_{BS, PFAS}$ values may also be the practical upper lifetime limit of biosolids application for these sites. Concentrations of PFAS of current regulatory concern (e.g., PFOA and PFOS) in municipal biosolids will vary between source municipalities and could ultimately diminish over time as these compounds are phased out of consumer products. Additionally, methods to treat the biosolids to reduce PFAS concentrations prior to land application should be evaluated. Any reduction of the PFAS mass loading will extend the beneficial and safe use of these products for agricultural applications.

The USEPA recently updated its lifetime drinking water health advisory limit for PFOA and PFOS to 0.004 ng/L and 0.02 ng/L, respectively (USEPA, 2022). However, at the time of this publication, it is unclear whether the PRG has also been updated, to what degree these recommended advisory levels will be adapted as state-level regulatory limits, and whether analytical methods will be developed that can routinely and reliably evaluate these levels in drinking water. Regardless, the approach and methods described in this current evaluation could prove useful to agricultural practitioners/management as these environmental regulatory metrics change in the future to reflect new toxicity data and regulation of additional PFAS.

4. Conclusions

Because PFAS are present in land-applied municipal biosolids, it is important to understand the fate and transport of these contaminants in the vadose zone to assess the potential for groundwater contamination. In this work, numerical simulations were employed to evaluate the significance of land-applied biosolids as a long-term source of PFAS in agricultural applications under natural infiltration conditions using flow and PFAS-specific transport parameters derived from existing site information or the literature. Simulations were performed for two test sites located in the same climatic zone with differing vadose zone hydrostratigraphy. Results of these simulations demonstrated the following:

- Within the biosolids-amended PFAS source, soil sorption/desorption processes were the dominant retention mechanisms for PFAS, with rate-limited desorption contributing minimally to the overall release rate of PFAS from the source. This result is significant because it suggests that over the long-term, rates of PFAS mass loss from the surficial biosolids-amended soil source are most dependent on the accurate characterization of the equilibrium soil interaction parameters than on those for non-equilibrium processes. This, in turn, simplifies leaching calculations.
- While AWI adsorption can be an important process for retaining PFAS in the vadose zone, this process did not contribute significantly to the overall release rate of PFAS from the biosolids-amended source soils at either test site. Climate conditions at these sites provided

persistently elevated soil moisture conditions and correspondingly lower A_{aw} values within these surface soils that minimized the contribution of AWI adsorption.

- Within the vadose zone soils underlying biosolids-amended source soils, AWI adsorption was observed to be most contributive to the retention of PFAS at Site K, where the vadose zone was characterized and modeled as a predominantly homogeneous vadose zone of silty sand. At Site K, lower moisture conditions, and thus larger A_{aw} values, were maintained throughout the 150-year simulation period. This condition allowed AWI adsorption to play a more significant role in limiting the rate of PFAS transport to groundwater, particularly for the more surface-active PFAS.
- In contrast, textural heterogeneities and finer-grained soils at Site W collectively minimized the role of AWI adsorption, resulting in faster transport to groundwater for the same PFAS source condition, even with an overall lower unsaturated hydraulic conductivity. Finer-grained soils comprising the upper vadose zone at Site W retained more moisture throughout the period of simulation, resulting in lower A_{aw} values and less overall retention of PFAS. These results are consistent with prior work highlighting the impact of vadose zone hydrostratigraphy on the significance of AWI adsorption as a mechanism for PFAS retention during leaching from a surficial source (Silva et al., 2020).
- Individual PFAS leach from a biosolids source, transport through the vadose zone at different rates, and reach the saturated zone at different times. For linear perfluoroalkyl PFAS employed herein, the rate of transport is dependent on the PFAS C_N , which impacts its surface activity and the interaction with the soil solid phase and its surface activity. For two of the PFAS of federal regulatory concern used here (i.e., PFOA and PFOS), peak concentrations arrive at the saturated zone many decades apart (Fig. 7). This requires treating PFOA and PFOS separately when applying USEPA guidance on PFAS limits for groundwater.
- Increasing the BLR at both sites increased PFAS concentrations in the leachate but did not impact the rate of release from the source zone. Increasing the number of annual biosolids application events also increased PFAS concentrations but did not significantly affect the rate of transport of individual PFAS through the vadose zone. However, it did increase the overall mass of PFAS leaching through the soil for a given BLR.
- Peak PFAS concentrations reaching the saturated zone increased linearly with increasing BLR and increased nonlinearly with increasing biosolids re-application at a given BLR. The results of these simulations were used to develop an empirical equation (Eq. (2)) that was, in turn, used to directly evaluate the impact of biosolids-amended soil PFAS loading rates and annual re-application periods on PFAS leaching concentrations for a given test site. This approach could be useful in developing site-specific soil screening levels for PFAS in support of BMPs for land application of biosolids containing PFAS at additional sites. Likewise, an up-scaled version of this approach could be used to develop state- or county-level soil screening levels for generic soil conditions most common in areas where biosolids land application sites are present.
- Example soil screening calculations for Site K (Table 2) also demonstrate that the numerical modeling approach used here can support the continued safe use of biosolids with respect to managing impacts related to PFAS and groundwater contamination in a manner consistent with the methods outlined in USEPA 40 CFR Part 503 for other biosolid-specific contaminants.

In this work, it was assumed that a finite mass of PFAS was applied as either a single application or as incremental additions associated with an annual biosolids application. However, recent investigations have shown the potential for degradation of PFAS precursor compounds within land-applied biosolids that could be represented as an additional source of some PFAS over the time-scales represented herein (Roehler

et al., 2021). For the current case, this additional source of PFAS could impact the magnitude of $C_{w, max}$ values and breakthrough profiles in Figs. 7 and 10 would plateau at a different $C_{w, max}$ for those PFAS being produced. Given that the rates and duration of PFAS production can be characterized, such a source of PFAS could be simulated as an in situ production process within the source zone, as was done in our prior work (Silva et al., 2020). However, simulated PFAS arrival times and the general approach proposed here to establish soil-screening levels and assist management of PFAS loading rates for biosolids application remain appropriate.

CRediT authorship contribution statement

Jeff A.K. Silva: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Jennifer L. Guelfo:** Methodology, Formal analysis, Resources, Writing – review & editing. **Jiří Šimůnek:** Methodology, Software, Writing – review & editing. **John E. McCray:** Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jconhyd.2022.104089>.

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